Payment Terminal - User Manual

Welcome to the Payment Terminal. This application serves as a live demonstration of Object-Oriented Programming (specifically Polymorphism), featuring a functional frontend interface dynamically connected to a MongoDB backend.

1. Installation & Setup

To run the application locally on your machine, follow these steps:

1. Open your terminal and navigate to the project directory.
2. Run `npm install` to install the required backend dependencies (Express, Mongoose, dotenv).
3. Ensure you have a `.env` file in the root directory containing your MongoDB connection string:
   `MONGO_URI=mongodb+srv://...`
4. Start the backend server by running: `node server.js`
5. Open your web browser and navigate to `http://localhost:3000` to view the UI.

2. Navigating the Interface

The interface is divided into two primary sections:

- **The Control Panel (Left):** Where you select payment types, input data, and manage the queue.
- **The Ledger (Right):** Where processed receipts are printed and mathematical totals are aggregated.

Step 1: Verify Connection

Before adding payments, check the bottom-left corner of the terminal window. A green dot with the text **"MongoDB connected"** indicates that the frontend is successfully communicating with the backend API and database.

Step 2: Queueing Payments

You can add three distinct types of payments to the system. The terminal uses a shared queue to hold them all simultaneously.

- **Credit Card:** Select the Credit Card tab. Enter an amount, currency, and valid card details. Click *"Add to queue"*.
- **PayPal:** Select the PayPal tab. Enter a valid email containing an '@' symbol and a password. Click *"Add to queue"*.
- **Bank Transfer:** Select the Bank Transfer tab. Enter account and routing numbers. Click *"Add to queue"*.

> **Demonstration Tip:** Use the *"Fill sample details"* button located on each tab to instantly populate the form with valid testing data during live presentations!

Step 3: Processing the Queue

Once you have added multiple payments to the queue, click the large **"Process all"** button.

- The system's Execution Engine will dynamically loop through the queue, identifying the true nature of each payment object and executing its specific processing logic (Dynamic Dispatch).
- The frontend securely transmits the processed data to the Express backend via HTTP POST.
- The backend successfully saves the transaction records permanently into MongoDB Atlas.

Step 4: Viewing the Ledger

Watch the right-side panel (the Ledger) after processing:

- Highly visual Receipt Tickets will automatically generate for each successful payment.
- The **Total Value** and **Processed Count** at the top of the ledger will mathematically aggregate and update in real-time.
